

- 4 (a) force = $q_1 q_2 / 4\pi\epsilon_0 x^2$ C1
 = $(6.4 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times \{12 \times 10^{-6}\}^2)$ C1
 = 2.56×10^{-17} N A1 [3]
- (b) potential at P is same as potential at Q B1
 work done = $q\Delta V$ M1
 $\Delta V = 0$ so zero work done A0 [2]
- (c) at midpoint, potential is $2 \times (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 6 \times 10^{-6})$ C1
 at P, potential is $(6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 3 \times 10^{-6}) + (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$ C1
 change in potential = $(6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$
 energy = $1.6 \times 10^{-19} \times (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$ C1
 = 1.0×10^{-22} J A1 [4]

5 (a) e.g. 'storage of charge' / storage of energy